Sahaj Bhadja

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Summary

Software engineer with 2+ years building full-stack applications, responsive UI/UX, real-time services, and mobile workflows. Experienced in C++, JavaScript, TypeScript, React, React Native, Node.js, REST APIs, algorithms, performance optimization, testing, debugging, and production deployment.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built Python APIs, Azure SQL services, and React dashboards for 500+ users, profiling indexed queries and backend access paths to reduce production latency from 1.2 seconds to 350 milliseconds.
- Orchestrated 5,000+ monthly jobs through explicit states, retries, reconciliation, and failure isolation, tripling throughput while improving API observability, operational resilience, recovery, maintainability, and systematic production issue diagnosis.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed React interfaces and Flask REST services handling 10,000+ daily requests, implementing responsive feature workflows, authentication, progress states, and reusable components while resolving production defects affecting 150+ users.
- Created multithreaded processing for 1,000+ files with validation, AWS RDS persistence, progress monitoring, and failure recovery, reducing turnaround 60% while maintaining responsive behavior under concurrent production load.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Developed Python services and scalable REST integrations using object-oriented patterns, filtering, pagination, caching, AWS Glue, S3, and SQL, improving production response times 30% and processing efficiency 68%.
- Maintained 20+ automated unit and integration suites with Jenkins, PyTest, and Mockito, accelerating technical troubleshooting, release verification, and reliable cloud application deployments across Windows, Linux, and macOS.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed modular React and Node.js features across eight microservices with authentication, role-based access, WebSocket messaging, and responsive interfaces, processing 2 million+ Azure SQL records through documented APIs.
- Automated Docker builds, functional tests, staged deployments, and production monitoring through Jenkins and Azure, reducing release time 50% while improving service reliability, deployment consistency, and operational visibility.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built React Native clients for iOS and Android with Firebase persistence, real-time state synchronization, image storage, push notifications, and secure sessions across a mobile platform handling 50,000+ records.
- Developed Flask REST APIs for bookings, pricing, and payments, delivering sub-200-millisecond responses and reliable webhook-driven synchronization that improved confirmation speed 40% across customer-facing interactive mobile application workflows.

Skills

Languages & Fundamentals: C++, JavaScript, TypeScript, Python, Java, SQL, object-oriented programming, data structures, algorithms, software design, performance analysis
Frontend & Mobile: React, React Native, Next.js, HTML5, CSS3, responsive UI/UX, reusable components, state synchronization, mobile applications, cross-device workflows
Backend & Data: Node.js, Flask, FastAPI, REST APIs, WebSockets, webhooks, microservices, Firebase, MongoDB, PostgreSQL, Azure SQL, AWS RDS, caching, authentication
Quality & Delivery: Git, Docker, Kubernetes, Jenkins, CI/CD, unit testing, integration testing, functional testing, automated testing, debugging, performance profiling, logging, monitoring, code review, documentation, Agile, Scrum

Projects

CPU-Scheduling Simulator | Live | *C++, React, JavaScript, Algorithms, Performance Analysis*
- Built a C++ simulator implementing six scheduling algorithms with deterministic workloads, edge-case validation, Gantt visualization, and performance analysis, reducing average turnaround time 25% across 1,000+ process configurations.

FinNexus | Live | *TypeScript, Next.js, REST APIs, WebSockets, MongoDB, Kubernetes*
- Built a responsive platform supporting 50,000+ users and one million transactions, integrating secure APIs, real-time WebSocket synchronization, MongoDB persistence, reusable interfaces, Kubernetes deployment, and automated release workflows.

PiXMod | Live | *Next.js, TypeScript, MongoDB, Stripe, CLIP, SAM*
- Built a full-stack AI application using Next.js, TypeScript, CLIP, and SAM, implementing reusable interfaces, structured image processing, secure authentication, MongoDB persistence, Stripe integration, and responsive cross-device workflows.